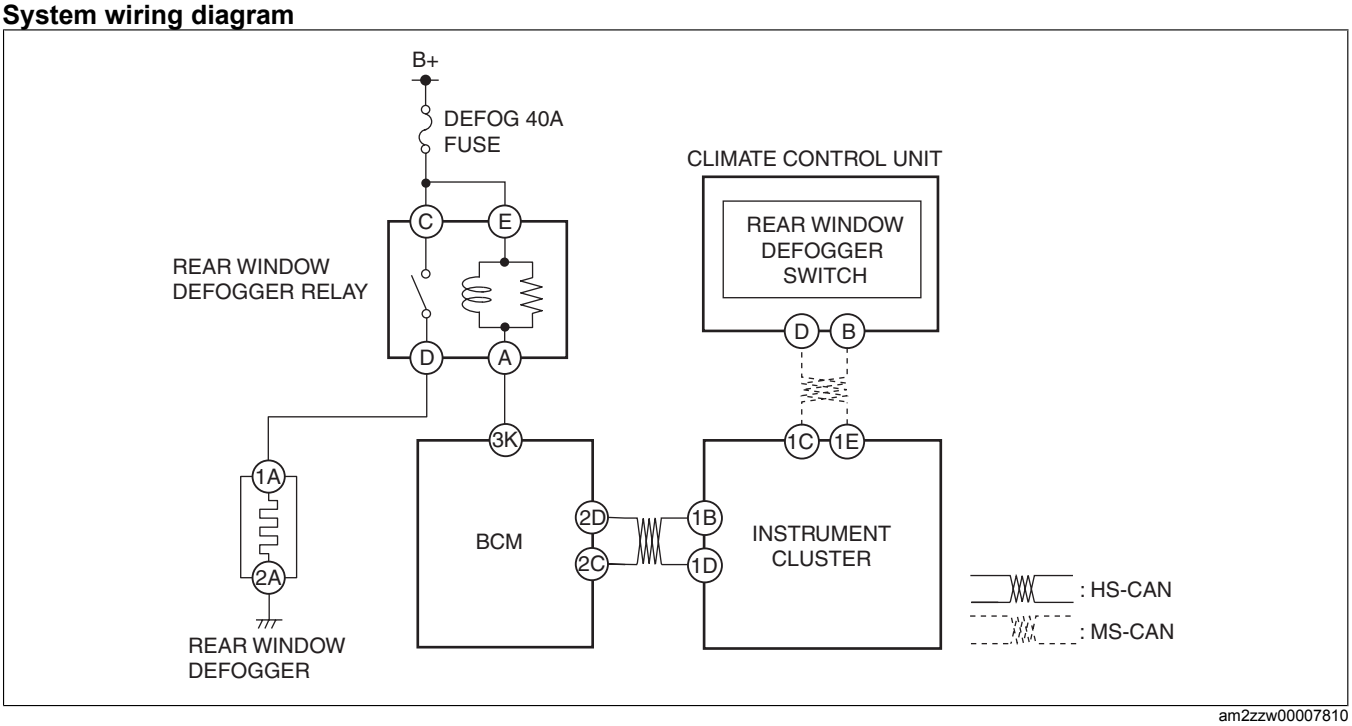
REAR WINDOW DEFOGGER SYSTEM INSPECTION

id091200020400

Full-auto Air Conditioner
System wiring diagram



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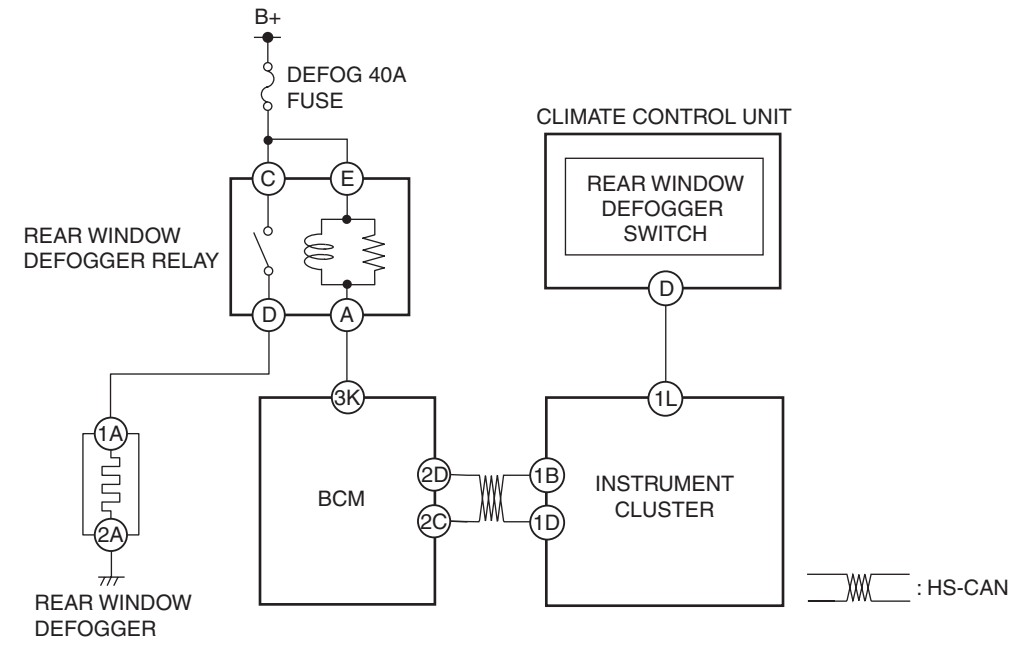
Inspection Procedure

Step	Inspection		Action
1	**VERIFY INSTRUMENT CLUSTER DTCs** • Switch the ignition ON (engine off or on). • Perform the DTC inspection for the instrument cluster using the M-MDS. (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Is a DTC displayed?	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	Go to the next step.
2	**DETERMINE MALFUNCTIONING LOCATION BY PERFORMING PID/DATA MONITOR INSPECTION** • Display the BCM PID [R_DEF_RLY] using the M-MDS. (See PID/DATA MONITOR INSPECTION [BCM].) • Turn the rear window defogger switch on. • Is the PID value on?	Yes	Go to Step 5.
		No	Go to the next step.
3	**INSPECT REAR WINDOW DEFOGGER SWITCH CONDITION BY PERFORMING PID/DATA MONITOR INSPECTION** • Display the climate control unit PID [R/DEF_SW] using the M-MDS. (See PID/DATA MONITOR DISPLAY [FULL-AUTO AIR CONDITIONER].) • Turn the rear window defogger switch on. • Is the PID value on?	Yes	Go to the next step.
		No	Replace the climate control unit, then go to Step 10. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
4	**INSPECT REAR WINDOW DEFOGGER OPERATION REQUEST SIGNAL FROM CLIMATE CONTROL UNIT BY PERFORMING PID/DATA MONITOR INSPECTION** • Display the climate control unit PID [R/DEF_CS] using the M-MDS. (See PID/DATA MONITOR DISPLAY [FULL-AUTO AIR CONDITIONER].) • Turn the rear window defogger switch on. • Is the PID value on?	Yes	**Note** • The rear window defogger does not operate normally due to BCM control malfunction. Replace the BCM, then go to Step 10. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Replace the climate control unit, then go to Step 10. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)

Step	Inspection		Action
5	**INSPECT REAR WINDOW DEFOGGER RELAY** • Switch the ignition OFF (LOCK). • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the rear window defogger relay. (See RELAY LOCATION.) • Inspect the rear window defogger relay. (See RELAY INSPECTION.) • Is the rear window defogger relay normal?	Yes	Go to the next step.
		No	Replace the rear window defogger relay, then go to Step 10. (See RELAY LOCATION.)
6	**INSPECT REAR WINDOW DEFOGGER RELAY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the rear window defogger relay is removed. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at rear window defogger relay terminals C and E (vehicle wiring harness side). • Is the voltage B+?	Yes	Go to the next step.
		No	Inspect the DEFOG 40 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between DEFOG 40 A fuse and rear window defogger relay terminals C and E. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify if there is a common connector between positive battery terminal and rear window defogger relay terminals C and E. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
7	**INSPECT REAR WINDOW DEFOGGER CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Install the rear window defogger relay. • Turn the rear window defogger switch on. • Measure the voltage at rear window defogger terminal 1A (vehicle wiring harness side). • Is the voltage B+?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify if there is a common connector between rear window defogger relay terminals D and rear window defogger terminal 1A. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground or open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground or has an open circuit. After repair procedure, go to Step 10.
8	**INSPECT REAR WINDOW DEFOGGER** • Inspect the rear window defogger. (See REAR WINDOW DEFOGGER INSPECTION.) • Is the rear window defogger normal?	Yes	Go to the next step.
		No	Repair the rear window defogger, then go to Step 10. (See REAR WINDOW DEFOGGER REPAIR.)

Step	Inspection		Action
9	**INSPECT BCM CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Switch the ignition ON (engine off or on). • Measure the voltage at BCM terminal 3K (vehicle wiring harness side). • Is the voltage B+?	Yes	Replace the BCM, then go to the next step. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Refer to the wiring diagram and verify if there is a common connector between BCM terminal 3K and rear window defogger relay terminal A. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground or open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground or has an open circuit. After repair, go to the next step.
10	**VERIFY IF MALFUNCTION IS RESOLVED** • Does the rear window defogger system operate normally?	Yes	Explain the contents of the servicing to the customer.
		No	If the malfunction has not been resolved, repeat the inspection from Step 1.

Manual Air Conditioner
System wiring diagram



am2zzw00007811

Inspection Procedure

Step	Inspection		Action
1	**VERIFY INSTRUMENT CLUSTER DTCs** • Switch the ignition ON (engine off or on). • Perform the DTC inspection for the instrument cluster using the M-MDS. (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Is a DTC displayed?	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	Go to the next step.
2	**DETERMINE MALFUNCTIONING LOCATION BY PERFORMING PID/DATA MONITOR INSPECTION** • Display the BCM PID [R_DEF_RLY] using the M-MDS. (See PID/DATA MONITOR INSPECTION [BCM].) • Turn the rear window defogger switch on. • Is the PID value on?	Yes	Go to Step 7.
		No	Go to the next step.

Step	Inspection		Action
3	**INSPECT REAR WINDOW DEFOGGER SWITCH CONDITION BY INSPECTING CLIMATE CONTROL UNIT** • Perform terminal voltage inspection at climate control unit terminal D. (See CLIMATE CONTROL UNIT INSPECTION [MANUAL AIR CONDITIONER].) • Is the voltage normal?	Yes	Go to the next step.
		No	Replace the climate control unit, then go to Step 12. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [MANUAL AIR CONDITIONER].)
4	**INSPECT CLIMATE CONTROL UNIT CIRCUIT FOR SHORT TO GROUND** • Disconnect the instrument cluster connector. • Verify that the climate control unit connector is disconnected. • Inspect for continuity between climate control unit terminal D (vehicle wiring harness side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify if there is a common connector between climate control unit terminal D and instrument cluster terminal 1L. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. After repair procedure, go to Step 12.
		No	Go to the next step.
5	**INSPECT CLIMATE CONTROL UNIT CIRCUIT FOR OPEN CIRCUIT** • Verify that the climate control unit connector and the instrument cluster connector are disconnected. • Inspect for continuity between climate control unit terminal D (wiring harness side) and instrument cluster terminal 1L (wiring harness side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify if there is a common connector between climate control unit terminal D and instrument cluster terminal 1L. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. After repair procedure, go to Step 12.
6	**INSPECT CLIMATE CONTROL UNIT CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the climate control unit connector and the instrument cluster connector are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Turn the rear window defogger switch on. • Measure the voltage at climate control unit terminal D (vehicle wiring harness side). • Is the voltage less than 1.0 V?	Yes	Replace the climate control unit, then go to Step 12. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [MANUAL AIR CONDITIONER].)
		No	Refer to the wiring diagram and verify if there is a common connector between climate control unit terminal D and instrument cluster terminal 1L. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to power supply to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to the power supply. After repair procedure, go to Step 12.
7	**INSPECT REAR WINDOW DEFOGGER RELAY** • Switch the ignition OFF (LOCK). • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the rear window defogger relay. (See RELAY LOCATION.) • Inspect the rear window defogger relay. (See RELAY INSPECTION.) • Is the rear window defogger relay normal?	Yes	Go to the next step.
		No	Replace the rear window defogger relay, then go to Step 12. (See RELAY LOCATION.)

Step	Inspection		Action
8	**INSPECT REAR WINDOW DEFOGGER RELAY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the rear window defogger relay is removed. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at rear window defogger relay terminals C and E (vehicle wiring harness side). • Is the voltage B+?	Yes	Go to the next step.
		No	Inspect the DEFOG 40 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify if there is a common connector between DEFOG 40 A fuse and rear window defogger relay terminals C and E. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify if there is a common connector between positive battery terminal and rear window defogger relay terminals C and E. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.
9	**INSPECT REAR WINDOW DEFOGGER CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Install the rear window defogger relay. • Turn the rear window defogger switch on. • Measure the voltage at rear window defogger terminal 1A (vehicle wiring harness side). • Is the voltage B+?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify if there is a common connector between rear window defogger relay terminals D and rear window defogger terminal 1A. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground or open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground or has an open circuit. After repair procedure, go to Step 12.
10	**INSPECT REAR WINDOW DEFOGGER** • Inspect the rear window defogger. (See REAR WINDOW DEFOGGER INSPECTION.) • Is the rear window defogger normal?	Yes	Go to the next step.
		No	Repair the rear window defogger, then go to Step 12. (See REAR WINDOW DEFOGGER REPAIR.)

Step	Inspection		Action
11	**INSPECT BCM CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Switch the ignition ON (engine off or on). • Measure the voltage at BCM terminal 3K (vehicle wiring harness side). • Is the voltage B+?	Yes	Replace the BCM, then go to the next step. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Refer to the wiring diagram and verify if there is a common connector between BCM terminal 3K and rear window defogger relay terminal A. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground or open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground or has an open circuit. After repair, go to the next step.
12	**VERIFY IF MALFUNCTION IS RESOLVED** • Does the rear window defogger system operate normally?	Yes	Explain the contents of the servicing to the customer.
		No	If the malfunction has not been resolved, repeat the inspection from Step 1.